

Networking Training Program (Module 6)

1. ARP

- ARP resolves IP addresses to MAC addresses, enabling devices to communicate within a local network.
- It allows routers to determine the next-hop MAC address for packet forwarding.
- ARP caching improves efficiency by reducing repeated address lookups.
- Without ARP, network devices wouldn't know where to send packets, disrupting communication.

The image shows a Wireshark packet capture window titled "Capturing from Ethernet3". The filter bar at the top is set to "arp". The packet list pane shows several ARP packets. Packet 210 is selected, and its details pane is expanded to show the "Address Resolution Protocol (request)" section. The packet bytes pane shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
3	0.044199	26:19:98:a2:0f:e0	52:c6:a7:b9:f6:7c	ARP	42	who has 192.168.164.234? Tell 192.168.164.132
4	0.044218	52:c6:a7:b9:f6:7c	26:19:98:a2:0f:e0	ARP	42	192.168.164.234 is at 52:c6:a7:b9:f6:7c
80	14.379883	26:19:98:a2:0f:e0	52:c6:a7:b9:f6:7c	ARP	42	who has 192.168.164.234? Tell 192.168.164.132
81	14.379904	52:c6:a7:b9:f6:7c	26:19:98:a2:0f:e0	ARP	42	192.168.164.234 is at 52:c6:a7:b9:f6:7c
210	28.715703	26:19:98:a2:0f:e0	52:c6:a7:b9:f6:7c	ARP	42	who has 192.168.164.234? Tell 192.168.164.132
211	28.715722	52:c6:a7:b9:f6:7c	26:19:98:a2:0f:e0	ARP	42	192.168.164.234 is at 52:c6:a7:b9:f6:7c
468	42.795524	26:19:98:a2:0f:e0	52:c6:a7:b9:f6:7c	ARP	42	who has 192.168.164.234? Tell 192.168.164.132
469	42.795539	52:c6:a7:b9:f6:7c	26:19:98:a2:0f:e0	ARP	42	192.168.164.234 is at 52:c6:a7:b9:f6:7c
500	46.133741	52:c6:a7:b9:f6:7c	26:19:98:a2:0f:e0	ARP	42	who has 192.168.164.132? Tell 192.168.164.234
501	46.136578	26:19:98:a2:0f:e0	52:c6:a7:b9:f6:7c	ARP	42	192.168.164.132 is at 26:19:98:a2:0f:e0
738	57.131391	26:19:98:a2:0f:e0	52:c6:a7:b9:f6:7c	ARP	42	who has 192.168.164.234? Tell 192.168.164.132
739	57.131425	52:c6:a7:b9:f6:7c	26:19:98:a2:0f:e0	ARP	42	192.168.164.234 is at 52:c6:a7:b9:f6:7c
818	73.258876	26:19:98:a2:0f:e0	52:c6:a7:b9:f6:7c	ARP	42	who has 192.168.164.234? Tell 192.168.164.132
819	73.258910	52:c6:a7:b9:f6:7c	26:19:98:a2:0f:e0	ARP	42	192.168.164.234 is at 52:c6:a7:b9:f6:7c

Frame 210: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{0C98CD43-4FBF-4A40-B98C-7CC48F20A5BE}, Ethernet II, Src: 26:19:98:a2:0f:e0 (26:19:98:a2:0f:e0), Dst: 52:c6:a7:b9:f6:7c (52:c6:a7:b9:f6:7c)

Destination: 52:c6:a7:b9:f6:7c (52:c6:a7:b9:f6:7c)

Source: 26:19:98:a2:0f:e0 (26:19:98:a2:0f:e0)

Type: ARP (0x0806)

[Stream index: 0]

Address Resolution Protocol (request)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: request (1)

Sender MAC address: 26:19:98:a2:0f:e0 (26:19:98:a2:0f:e0)

Sender IP address: 192.168.164.132

Target MAC address: 00:00:00:00:00:00 (00:00:00:00:00:00)

Target IP address: 192.168.164.234

0000 52 c6 a7 b9 f6 7c 26 19 98 a2 0f e0 08 06 00 01 R... |&...
0010 00 00 06 04 00 01 26 19 98 a2 0f e0 c0 a8 a4 84&...
0020 00 00 00 00 00 00 c0 a8 a4 ea&...>

2. Manually configure and Traceroute Tracking

PC4

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

Interface

FastEthernet0

▼

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.16.1.10

Subnet Mask

255.255.255.0

Default Gateway

192.16.1.1

DNS Server

0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::250:FFF:FE37:D3EB

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MDS

▼

Username

Password

☐ Top

PC5

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.16.2.10

Subnet Mask 255.255.255.0

Default Gateway 192.16.2.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::203:E4FF:FE29:6075

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Top

```

Router#
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.16.1.1 255.255.255.0
Router(config-if)#
Router(config-if)#
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet1/0
%Invalid interface type and number
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 192.16.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#

```

```
Would you like to enter the initial configuration dialog? [yes/no]:
```

```
Press RETURN to get started!
```

```
Router>
Router>
Router>
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#interface gigabitethernet0/0
Router(config-if)#
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-S-CHANGED: Interface GigabitEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#exit
Router(config)#interface gigabitethernet0/1
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-S-CHANGED: Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#
```

```
Router#
Router#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.1.1	YES	manual	up	up
GigabitEthernet0/1	192.168.2.1	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

```
Router#
```

```
C:\>tracert 192.168.2.10
Invalid Command.
```

```
C:\>tracert 192.168.2.10
```

```
Tracing route to 192.168.2.10 over a maximum of 30 hops:
```

1	0 ms	0 ms	0 ms	192.168.1.1
2	0 ms	0 ms	0 ms	192.168.2.10

```
Trace complete.
```

3. Network Subnetting

Subnetting 10.0.0.0/24 into 4 Equal Subnets

1. Given Network Information

- Network Address: 10.0.0.0/24
- Subnet Mask: 255.255.255.0
- Total Hosts in /24: $2^8 - 2 = 254$ hosts
- Requirement: Divide into 4 equal subnets

2. Calculate the New Subnet Mask

- To divide into 4 subnets, we need 2 extra bits ($2^2 = 4$).
- The new subnet mask becomes /26 (255.255.255.192).
- Each subnet has $2^6 - 2 = 62$ usable hosts.

3. Subnet Allocation and Host Ranges

Subnet	Network Address	First Usable IP	Last Usable IP	Broadcast Address
1	10.0.0.0/26	10.0.0.1	10.0.0.62	10.0.0.63
2	10.0.0.64/26	10.0.0.65	10.0.0.126	10.0.0.127
3	10.0.0.128/26	10.0.0.129	10.0.0.190	10.0.0.191
4	10.0.0.192/26	10.0.0.193	10.0.0.254	10.0.0.255

4. Assign IP Addresses to Devices in Packet Tracer

- Router (Default Gateway): Assign first usable IP from each subnet.
- PCs, Switches: Assign IPs within the valid host range.

Example Device IP Assignments

Device	Assigned IP	Subnet Mask	Gateway
Router	10.0.0.1	255.255.255.192	N/A
PC1	10.0.0.2	255.255.255.192	10.0.0.1
PC2	10.0.0.65	255.255.255.192	10.0.0.64
PC3	10.0.0.129	255.255.255.192	10.0.0.128

Device	Assigned IP	Subnet Mask	Gateway
PC4	10.0.0.193	255.255.255.192	10.0.0.192

5. Verify Connectivity

1. Ping Test:

- Use the ping command to check connectivity between devices in the same and different subnets.
- Example:

ping 10.0.0.65

2. Traceroute Test:

- Use tracert or traceroute to check packet paths between different subnets.
- Example:

tracert 10.0.0.129

3. Check Routing Table (On Router):

4. show ip route

4. IP Address Classification and NAT Explanation

1. Identifying the Class of Each IP Address

IP addresses are categorized into different classes based on their first octet.

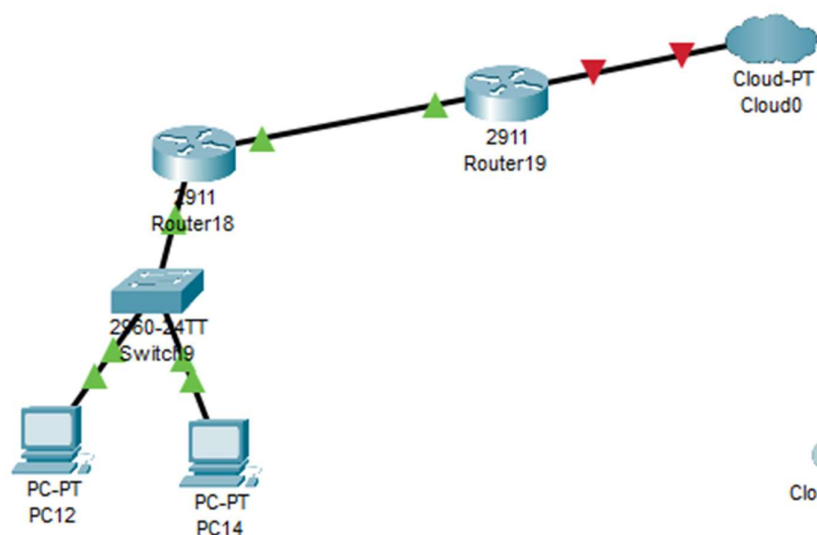
IP Address	First Octet Range	Class
192.168.10.5	192 - 223	Class C
172.20.15.1	128 - 191	Class B
8.8.8.8	1 - 126	Class A

2. Determining if the IP is Private or Public

Some IP ranges are reserved for private networks, while others are public and routable on the internet.

IP Address	Private/Public	Reason
192.168.10.5	Private	Falls in 192.168.0.0 - 192.168.255.255 (Private Class C)
172.20.15.1	Private	Falls in 172.16.0.0 - 172.31.255.255 (Private Class B)
8.8.8.8	Public	Google DNS, routable on the internet

5. NAT Translation



```
Router18
Physical Config CLI Attributes
IOS Command Line Interface

Router>
Router>
Router>
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

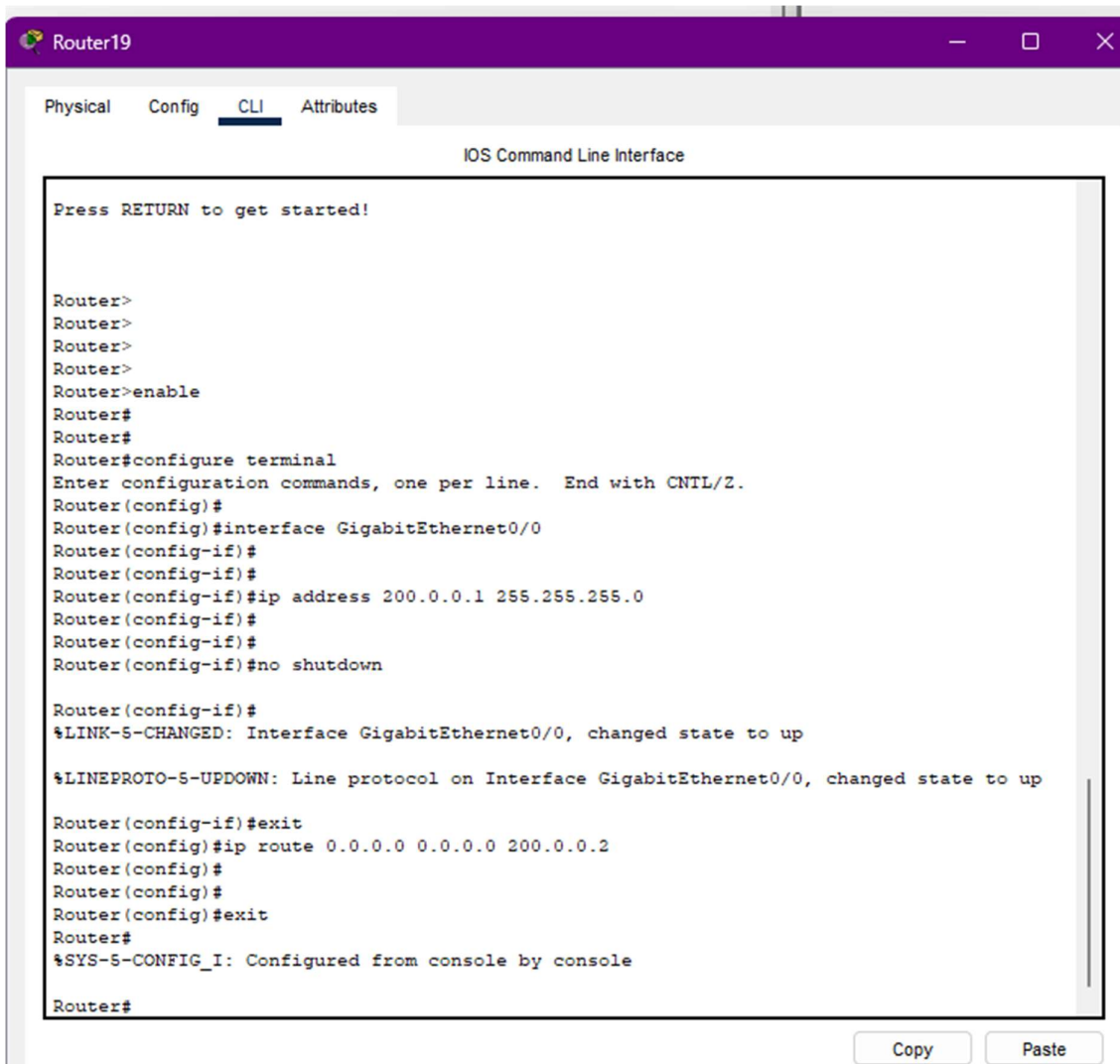
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

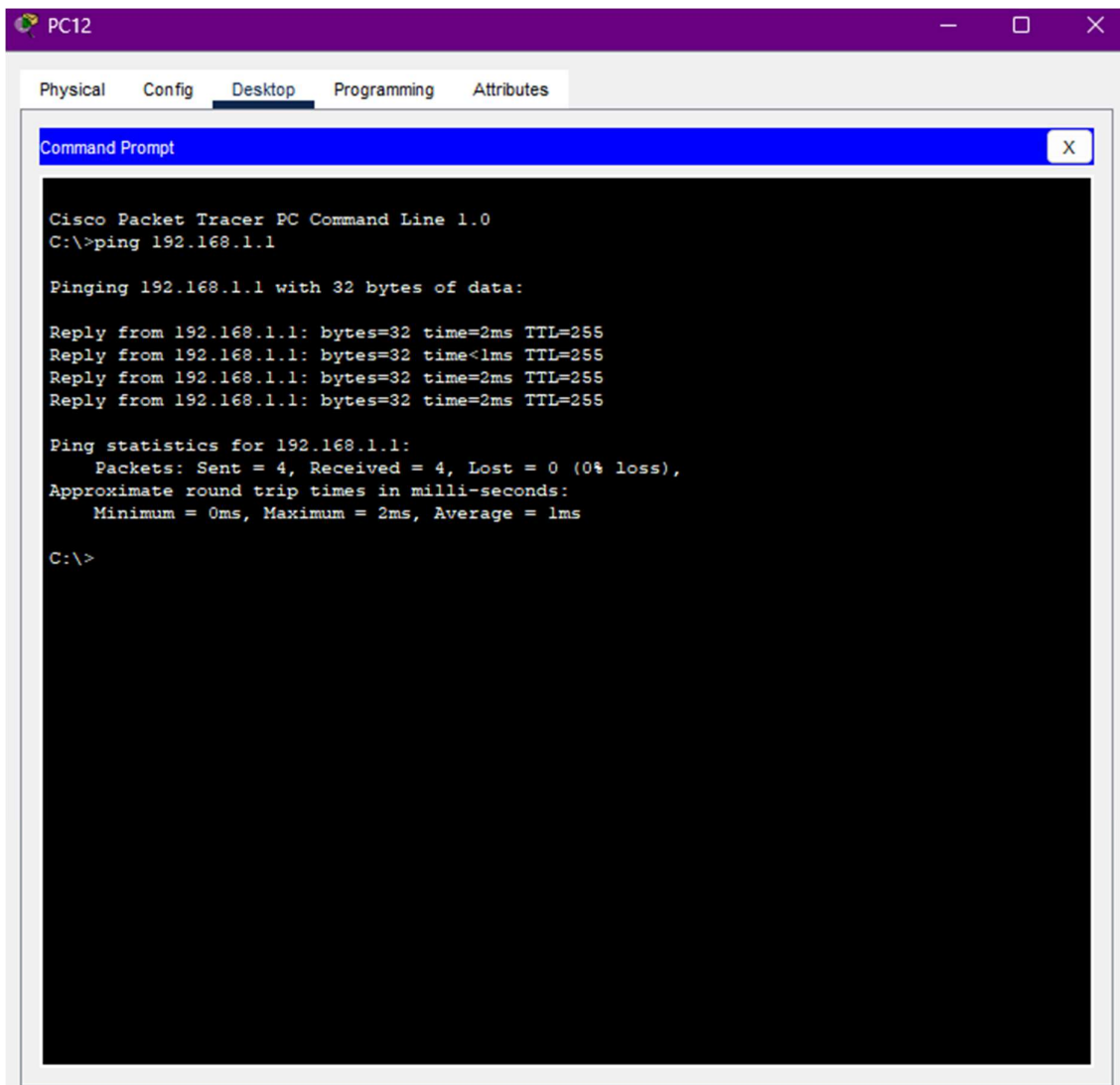
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#ip address 200.0.0.2 255.255.255.0
Router(config-if)#
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#
```

Copy Paste





```
IP NAT debugging is on
Router#
Router#
Router#
Router#
Router#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0
Hits: 3 Misses: 4
Expired translations: 4
Dynamic mappings:
Router#
```

```
Router#
Router#
Router#
NAT: s=192.168.1.20->203.0.113.2, d=203.0.113.10 [17]

NAT*: s=203.0.113.10, d=203.0.113.2->192.168.1.20 [4]

NAT: s=192.168.1.20->203.0.113.2, d=203.0.113.10 [18]

NAT*: s=203.0.113.10, d=203.0.113.2->192.168.1.20 [5]

NAT: s=192.168.1.20->203.0.113.2, d=203.0.113.10 [19]

NAT*: s=203.0.113.10, d=203.0.113.2->192.168.1.20 [6]

NAT: s=192.168.1.20->203.0.113.2, d=203.0.113.10 [20]

NAT*: s=203.0.113.10, d=203.0.113.2->192.168.1.20 [7]
```